

Reflective Journal

This assignment gave me a much clearer picture of what actually happens behind the scenes when an artificial intelligence model is prepared for deployment on an edge device. Before completing this lab, I mostly associated machine learning with training models and measuring accuracy. I did not fully realize how many additional steps are required to make a model usable outside of a traditional development environment. Working through the conceptual workflow, from setting up the environment to converting and deploying a model with TensorFlow Lite, helped me understand that deployment is just as important as training.

One of the main challenges I faced was trying to understand the idea of model conversion. At first, it seemed unnecessary to convert a trained TensorFlow model into another format. Through research, I learned that models created for training are not automatically optimized for real-world devices. Edge devices have limited memory, processing power, and energy, which means models must be smaller and more efficient. Understanding this difference changed how I view model design. It is not only about achieving high accuracy, but also about creating a model that can realistically run on the target device.

Another challenge was understanding the purpose of the TensorFlow Lite Interpreter. In typical programming projects, models are loaded and run with very few extra steps. In edge deployment, however, there is a more structured process: loading the model, allocating tensors, identifying input and output details, and carefully formatting data. Stepping through these actions helped me see how close edge AI operates to the hardware level. This made me realize why precision and attention to detail are critical when deploying models in production environments.

This assignment also helped reinforce my understanding of neural network architecture. Designing a simple model for the MNIST dataset made it clear that even small design choices, such as the number of neurons or layers, can affect both performance and efficiency. I learned that simpler models are often preferred for edge devices because they require fewer resources and run faster. This perspective will influence how I approach future projects, especially when working with environments that have limited capabilities.

A major learning outcome from this lab was recognizing the importance of data preprocessing. Normalizing pixel values and making sure consistent input shapes may seem like small steps, but they directly impact how well a model learns and performs. Thinking through these processes conceptually helped me understand why models can fail even when the architecture appears correct. Proper data preparation is just as essential as model design.

I also gained experience reading technical documentation more carefully. Since this was a conceptual assignment, I relied heavily on official TensorFlow and TensorFlow Lite resources to

understand how each step works. This pushed me to slow down and focus on understanding terminology and workflows rather than just copying instructions. Being able to understand documentation is a valuable skill that will continue to help me as I work with new technologies.

From a real-world perspective, this assignment made edge AI feel much more relevant and practical. Many everyday technologies, such as smartphones, smart cameras, voice assistants, and wearable devices, use edge AI to operate quickly and privately without relying on constant cloud access. Knowing how models are converted and deployed to run on these devices helps me better understand how these systems function behind the scenes.

Looking ahead, I feel more prepared for hands-on implementation in future labs. Because I now understand the purpose of each step, I will be able to approach practical deployment with more confidence. Instead of just following instructions, I will have a mental framework for what should happen and why. This will make it easier to troubleshoot issues and understand errors when they occur.

This assignment strengthened my understanding of the full lifecycle of an AI model, from start design to final deployment. It showed me that successful AI systems are not only accurate but also efficient, reliable, and thoughtfully engineered. This experience has given me a strong understanding that I can build on as I continue learning about artificial intelligence and edge computing.